SYNOPSIS

Project Group No: 26SOCUG1087

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Project Title : Detection of Uterine Fibroids using EfficientNetB0 and MobileNetV2

Name of the Guide: Dr.MJ.Jeyasheela Rakkini, Assistant Professor-II/SOC

ABSTRACT:

Uterine fibroids are one of the most common yet often overlooked health problems affecting women over 30, silently causing pelvic pain, irregular periods, and fertility struggles that can deeply affect everyday life. Catching them early makes a huge difference, but the reality is that reading ultrasound images by hand is slow, inconsistent, and easy to get wrong especially since fibroids can look a lot like other uterine conditions on a scan. That's exactly the gap this project set out to fill. We built an automated system powered by deep learning that can look at an ultrasound image and determine whether a fibroid is present, without needing a doctor to manually interpret every scan. The model at its core, EfficientNetB0, was enhanced with an attention mechanism that teaches it to zero in on the most meaningful parts of an image, much like how a trained radiologist knows where to look. We worked with 3,050 ultrasound images, evenly split between fibroid and healthy cases, and used data augmentation to prepare the model for the kind of variability seen in real clinical settings. Comparing two different models,EfficientNetB0 attained accuracy of 69.92% and MobileNetV2 stood out with a remarkable validation accuracy of nearly 98.96%, showing that a well-optimized lightweight model can actually outshine more complex architectures when the dataset is limited. At its heart, this project is about making early diagnosis more accessible and reliable giving doctors a smarter, faster tool to detect uterine fibroids with greater confidence and consistency.

Specific Contribution:

- Searching codes and implementing logical coding

Specific Learning:

- Gained hands-on experience in building and fine-tuning deep learning models for medical image detection using EfficientNetB0 and MobileNetV2.

Technical Limitations and Ethical Challenges faced:

- EfficientNetB0 showed moderate performance on the current dataset size, requiring larger data for optimal results.
- Attention mechanism improves focus but does not provide full explainability.

Keywords: *Uterine fibroid, Deep learning, EfficientNetB0, Attention mechanism, MobileNetV2*

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